

Quantifying oscillation & complexity of sea waves from video

A method reference for the `HNA.modules.video` module

HumanNatureAttunement project
Antoine Bellemare & Mar Estarellas

May 7, 2026

Abstract

The `HNA.modules.video` module turns a video clip into a battery of 1-D signals – sampled at the frame rate – that quantify both the *oscillation* (mean luminance, optical-flow magnitude, modal coefficients, timestack period) and the *complexity* (multiple entropies, fractal dimensions, Hjorth parameters, DFA exponent, $1/f$ slope) of natural visual phenomena such as sea waves. These signals are designed as drop-in inputs to the existing `HNA.modules.coupling` pipeline, which already estimates audio–EEG coupling for the project; the same statistics (windowed cross-correlation, magnitude-squared coherence, phase-locking value, weighted phase-lag index, mutual information) therefore extend symmetrically to the visual modality and to coupling with HRV / respiration / EMG. This report documents every method, gives the math, the chosen hyperparameters, and a worked example on three test clips.

This document is generated from the source at `report/main.tex` and compiles with `tectonic`. Method sources live under `src/HNA/modules/video.py`; the corresponding animation is under `remotion/`.

Contents

1	Motivation & design	2
2	Module overview	2
3	Methods	3
3.1	Global temporal envelope	3
3.2	Optical flow (Farneback)	3
3.3	Multi-scale optical flow	4
3.4	Per-frame spatial complexity	4
3.5	Per-frame 2D Fourier descriptors	6
3.6	Per-pixel temporal spectrum	7
3.7	Spatio-temporal modal decomposition (DMD)	8
3.8	Timestack analysis	10
3.9	Nonlinear-dynamics complexity summary	10
3.10	Time-resolved (windowed) complexity	10
4	Hyperparameters & defaults	11
5	Coupling integration	11
6	Validation	12
7	Reproducibility	13


```

6
7 # feats.signals          : Dict[str, np.ndarray]  (1-D, fs = fps)
8 # feats.complexity[key]  : Dict[str, float]      (15 measures)
9 # feats.timestack        : TimestackResult
10 # feats.modal            : ModalResult
11 # feats.meta             : VideoMeta

```

The high-level pipeline calls, in order, `extract_global_signals` $\rightarrow$ `extract_optical_flow_signals` $\rightarrow$ `extract_spatial_complexity` $\rightarrow$ `extract_timestack` $\rightarrow$ `extract_modal` $\rightarrow$ `complexity_summary`, with an optional `windowed_complexity` pass on a chosen subset of signals. Every produced 1-D signal is a valid input to the coupling tools (see Section 5).

3 Methods

Throughout, $I_t(x, y)$ denotes the grayscale frame at time $t \in \{1, \dots, N\}$ (sampled at $f_s = \text{fps}/\text{stride Hz}$), $g_t(x, y)$ its 192-px-on-the-long-axis downscaled copy, and $\mathbf{v}_t(x, y) = (u_t(x, y), v_t(x, y))$ the dense optical-flow field between g_{t-1} and g_t .

3.1 Global temporal envelope

Each frame is collapsed to a scalar by spatial averaging or by histogram statistics. These signals are the visual analogue of the Hilbert-derived acoustic envelope already used in the audio pipeline.

Per-frame scalars.

$$\text{lum}_t = \frac{1}{HW} \sum_{x,y} g_t(x, y), \quad (1)$$

$$\mathbf{R}_t, \mathbf{G}_t, \mathbf{B}_t = \text{channel means}, \quad (2)$$

$$\sigma_t = \text{std}_{x,y} g_t(x, y), \quad (3)$$

$$H_t^{\text{spat}} = -\sum_b p_t^{(b)} \log_2 p_t^{(b)}, \quad (4)$$

$$d_t = \frac{1}{HW} \sum_{x,y} |g_t(x, y) - g_{t-1}(x, y)|, \quad (5)$$

where $p_t^{(b)}$ is the normalized histogram of g_t in $B = 64$ bins. The last quantity, the *frame-difference* d_t , is a frame-rate motion energy proxy that already captures the swell rhythm in many recordings. $d_1 = 0$ by convention.

Why blue. The per-channel means are reported separately because the blue channel of sea-wave footage is unusually clean: it varies primarily with sky / water reflectance and tracks the swell envelope nearly as well as optical-flow magnitude, at a fraction of the cost.

3.2 Optical flow (Farneback)

Dense flow is estimated by the polynomial-expansion algorithm of [Farneback \(2003\)](#), implemented in OpenCV as `cv2.calcOpticalFlowFarneback`. Hyperparameters (Section 4) are tuned for textured water: `winsize = 21`, `levels = 4`, `iterations = 5`, `poly_n = 7`, `poly_sigma = 1.5`. From the field $\mathbf{v}_t$ we derive

$$\text{mag}_t = \frac{1}{HW} \sum_{x,y} \|\mathbf{v}_t(x, y)\|, \quad (6)$$

$$\text{mag}_t^{p95} = \text{percentile}_{95} \|\mathbf{v}_t\|, \quad (7)$$

$$\text{curl}_t = \frac{1}{HW} \sum_{x,y} |\partial_x v_t - \partial_y u_t|, \quad (8)$$

$$\text{div}_t = \frac{1}{HW} \sum_{x,y} (\partial_x u_t + \partial_y v_t), \quad (9)$$

$$\bar{u}_t, \bar{v}_t = \text{spatial means of } u, v, \quad (10)$$

$$H_t^{\text{dir}} = - \sum_k q_t^{(k)} \log_2 q_t^{(k)}, \quad (11)$$

with $q_t^{(k)}$ the histogram of flow directions $\arctan_2(v_t, u_t)$ over pixels with $\|\mathbf{v}_t\| > 0.1$, in $K = 16$ bins. The mean magnitude tracks bulk wave motion; mean $|\text{curl}|$ isolates rotational and turbulent content (foam break-up); the divergence is small in incompressible-flow regimes (water surface) but spikes near surf-zone breaking. Direction entropy quantifies how isotropic the motion field is at frame t .

3.3 Multi-scale optical flow

`extract_optical_flow_multiscale` returns the same family of descriptors at multiple granularities. Two granularity dimensions are exposed, both standard generalisations of pyramidal flow (Bouquet, 2001; Zach *et al.*, 2007):

Temporal stride. For each Δt in `temporal_strides` (default $\{1, 3, 9\}$ frames) we compute the flow between frames t and $t - \Delta t$. $\Delta t = 1$ captures ripple- and foam-scale motion; $\Delta t = 9$ at 24 fps probes ≈ 0.4 s motion – close to a quarter swell period. Because each Δt samples a different temporal band of the underlying motion field, these are essentially three independent flow channels rather than redundant copies of the same signal.

Spatial decomposition. After computing (u, v) at a given Δt , the field is split into a Gaussian low-pass component (default $\sigma = 6$ px) and the residual:

$$\mathbf{v}_{\text{large}} = G_\sigma * \mathbf{v}, \quad \mathbf{v}_{\text{small}} = \mathbf{v} - \mathbf{v}_{\text{large}}. \quad (12)$$

The large-scale component carries swell-direction/speed; the small-scale residual carries turbulent motion. The same four scalars (`mag_mean`, `curl_abs_mean`, `div_mean`, `dir_entropy`) are then computed on each of `full`, `large`, `small` fields.

Naming. Output keys follow the schema `flow_dt{N}_{full|large|small}_{stat}`, e.g. `flow_dt9_large_mag_mean`. With the default settings the function emits $|\Delta t| \times 3 \times 4 = 36$ frame-rate signals, each individually coupling-ready against EEG / HRV / EMG.

3.4 Per-frame spatial complexity

Five complementary descriptors of *within-frame* structure:

Edge density. Apply a Canny edge detector (Canny, 1986) with thresholds (35, 110) – tuned downward from OpenCV defaults to capture soft foam contours that 60/150 misses. Edge density is the fraction of pixels above threshold: $\rho_t = \frac{1}{HW} \sum_{x,y} \mathbf{1}[\text{edge}_t(x, y)]$.

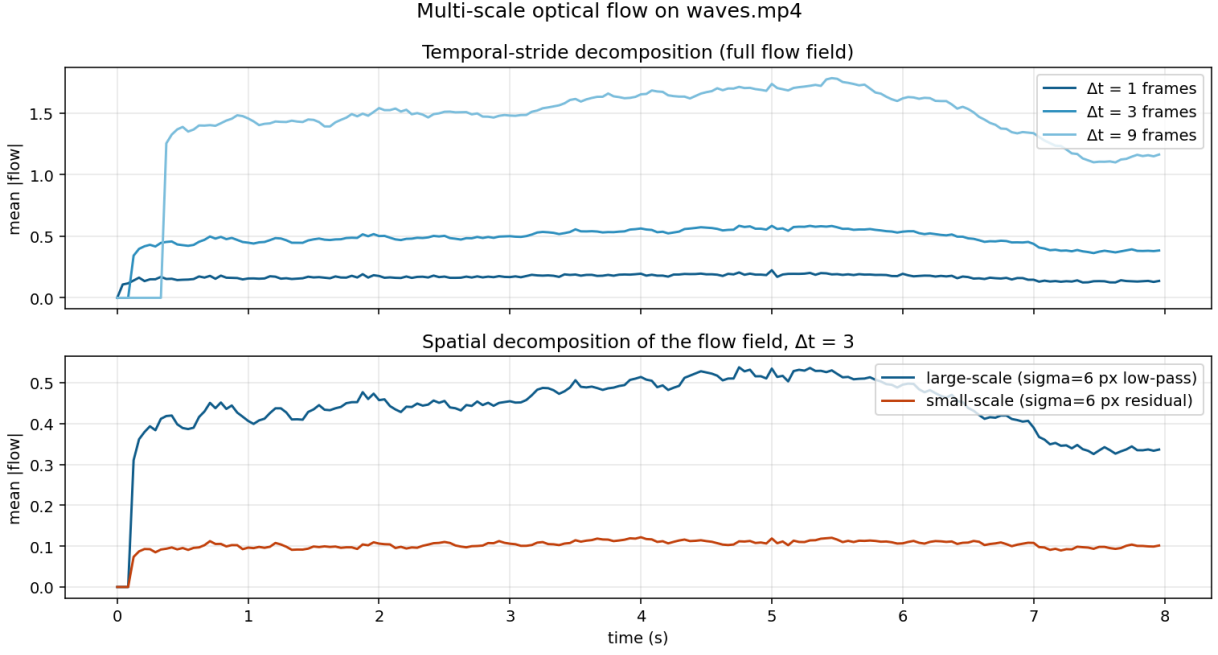


Figure 2: Multi-scale optical flow on `waves.mp4`. Top: temporal-stride decomposition of the full flow field – mean $|\mathbf{v}|$ scales roughly linearly with Δt (the longer- baseline flow integrates more swell motion). Bottom: spatial low-pass vs. residual split at $\Delta t = 3$. The large-scale component carries $\sim 80\%$ of the total motion energy (swell); the small- scale residual is the remaining turbulent content.

Radial 2D-FFT slope. The radial average of the 2-D power spectrum of g_t typically follows a power law $P(k) \propto k^{-\alpha}$ over a mid-frequency band. We fit

$$\log P(k) = -\alpha \log k + c \quad (13)$$

on $0.05 k_{\max} \leq k \leq 0.5 k_{\max}$ where $k_{\max} = \min(H, W)/2$. Sea / foam textures give $\alpha \in [1.5, 3]$. This is conceptually parallel to the EEG *aperiodic* ($1/f$) exponent in the spectral domain.

Box-counting fractal dimension (two binarisations). On a binary mask B , count non-empty boxes $N(\varepsilon)$ at sizes $\varepsilon \in \{2, 3, 4, 6, 8, 12, 16\}$. Fit

$$\log N(\varepsilon) = D \log(1/\varepsilon) + c, \quad (14)$$

the slope D is the box-counting fractal dimension. We apply this to two complementary binarisations of each frame: `fractal_dim_edges` on the Canny edge map (sharp foam contours) and `fractal_dim_grad` on the binarised gradient-magnitude image $|\nabla g_t| > \text{percentile}_{60}$, which captures the graded turbulent structure that the binary edge map throws away. For natural foam, $D \in (1.5, 2.0)$ depending on the binarisation.

Lacunarity. Box-counting answers *how rough*; lacunarity (Plotnick *et al.*, 1993; Allgood, 1991) answers *how clumpy / heterogeneous that roughness is at multiple scales* – two textures with the same D can have very different lacunarity. For each box size ε ,

$$\Lambda(\varepsilon) = 1 + \frac{\text{Var } N(\varepsilon)}{(\mathbb{E} N(\varepsilon))^2}, \quad (15)$$

and we report $\log \bar{\Lambda}$ on the same two binarisations (`lacunarity_edges`, `lacunarity_grad`). Fig. 3 shows that D alone does not separate the three example clips, while lacunarity on the gradient image clearly splits them.

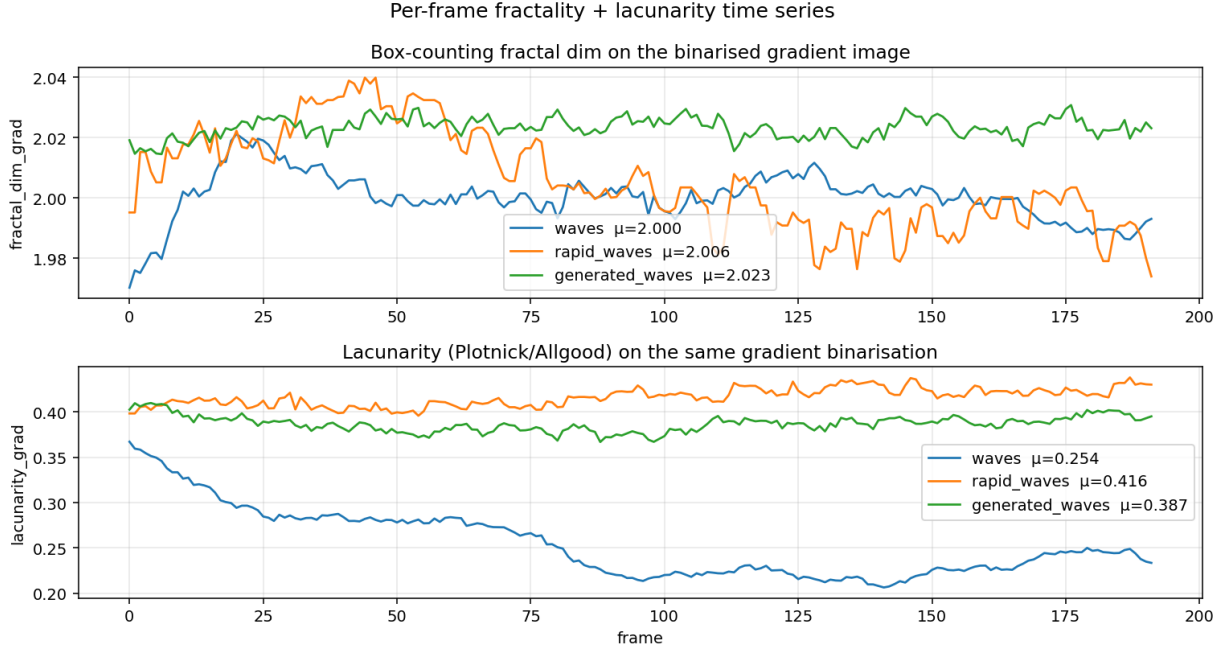


Figure 3: Per-frame fractality + lacunarity on the binarised gradient image, across the three example clips. Box-counting D (top) is nearly identical for the three sea states; lacunarity (bottom) separates them strongly – which is the whole reason it complements fractal dimension.

Patch Shannon entropy. Tile each frame into 24×24 blocks; compute the Shannon entropy of the intensity histogram inside every block; average. This produces a per-frame scalar *and* a low-resolution map (Fig. 1, rightmost panel) that is rendered as the `patch_heatmap.mp4` overlay.

GLCM contrast / homogeneity. Quantize g_t to $L = 16$ levels; build the gray-level co-occurrence matrix G_d at four offsets $d \in \{(0, 1), (1, 0), (1, 1), (1, -1)\}$; report contrast $= \sum_{ij} (i - j)^2 G_d(i, j)$ and homogeneity $= \sum_{ij} G_d(i, j) / (1 + (i - j)^2)$ (Haralick *et al.*, 1973). Implemented from scratch (no `scikit-image` dependency).

3.5 Per-frame 2D Fourier descriptors

The radial PSD slope (Sec. 3.4) collapses the 2-D spectrum to a single number; `extract_spatial_fft_signals` adds three richer per-frame scalars that exploit the full 2-D spectrum.

Let $G_t(k_x, k_y) = \mathcal{F}_{2D}\{g_t - \bar{g}_t\}$ and $P_t(k_x, k_y) = |G_t|^2$. With $r = \sqrt{k_x^2 + k_y^2}$ and ring mask $\mathcal{R} = \{2 \leq r \leq 0.95 r_{\max}\}$:

$$k_t^{\text{peak}} = \frac{r(\arg \max_{\mathcal{R}} P_t)}{r_{\max}}, \quad (16)$$

$$A_t = \log \frac{\sum_{\mathcal{R} \cap \angle \approx \pm \pi/2} P_t}{\sum_{\mathcal{R} \cap \angle \approx 0, \pi} P_t}, \quad (17)$$

$$\theta_t = \frac{1}{2} \arctan_2 \left(\sum_{\mathcal{R}} \sin(2\angle) P_t, \sum_{\mathcal{R}} \cos(2\angle) P_t \right), \quad (18)$$

where $\angle$ is the polar angle $\arctan_2(k_y, k_x)$. The peak wavenumber k_t^{peak} is the normalized dominant spatial frequency of foam / ripple texture; the *anisotropy* A_t is positive when horizontal stripes dominate (Fourier power on the k_y axis) and negative when vertical stripes dominate; $\theta_t \in [0, \pi)$ is the weighted-mean orientation. On `waves.mp4` we measure $A_t \approx 2.2$ on average –

consistent with the visible horizontal swell crests – and $\theta_t \approx \pi/2$, confirming that interpretation (Fig. 4).

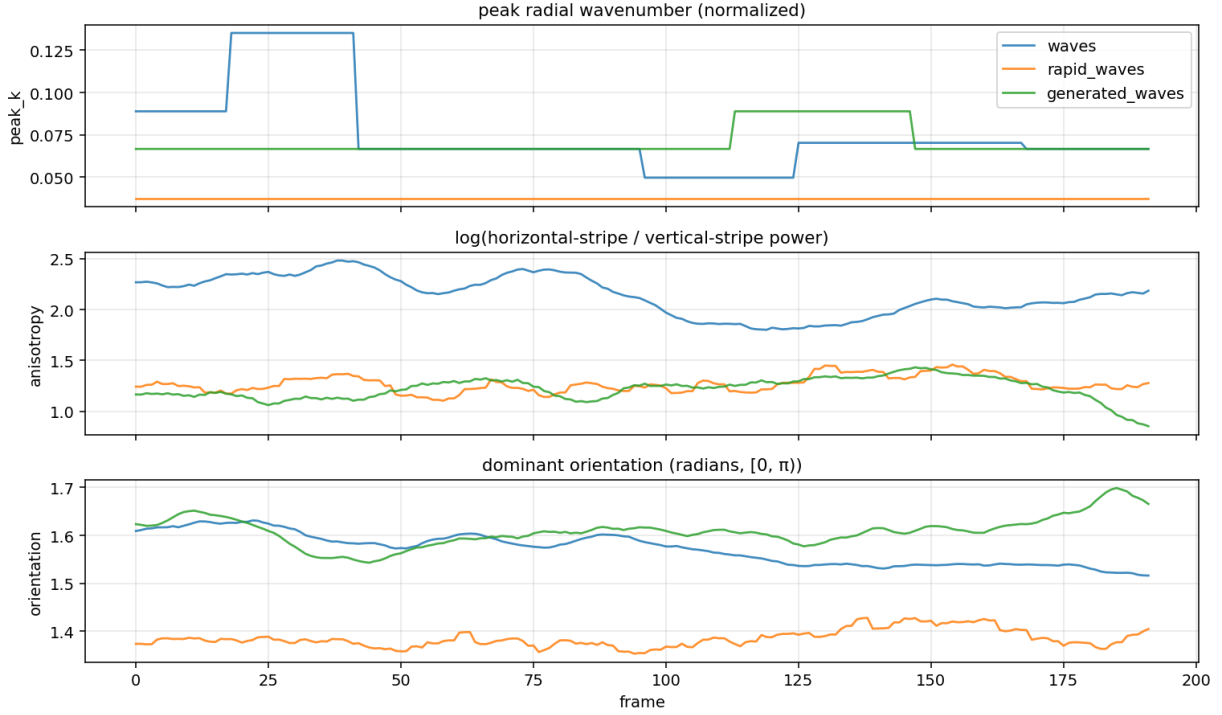


Figure 4: Per-frame 2-D-FFT descriptors across the three example clips. All natural-wave footage has positive anisotropy (horizontal swell crests dominate); the synthetic clip has the smallest peak wavenumber because it was generated with very low spatial frequency content.

3.6 Per-pixel temporal spectrum

So far every method either pooled in *space* (returning a 1-D signal in t) or in *time* (returning a 2-D image at a single t). The function `extract_pixel_spectrum` addresses the obvious missing direction: it computes, for every pixel (x, y) of a downsampled grayscale copy, the FFT of luminance over time, and reports the resulting (frequency $\times$ space) information as a small set of 2-D maps and global scalars.

Procedure. With a $T \times h \times w$ stack $X(t, y, x) = g_t(x, y)$:

1. Per-pixel linear detrend along t .
2. Hann taper w_t , then real-input FFT along t : $F(f, y, x) = \text{rfft}_t[w_t X(t, y, x)]$.
3. Power $\hat{P}(f, y, x) = |F(f, y, x)|^2$, ignoring DC.

Per-pixel maps. Each is a 2-D image of size (h, w) :

$$f^{\text{peak}}(y, x) = \arg \max_{f > 0} \hat{P}(f, y, x), \quad (19)$$

$$P^{\text{peak}}(y, x) = \max_{f > 0} \hat{P}(f, y, x), \quad (20)$$

$$H^{\text{spec}}(y, x) = - \sum_{f > 0} \tilde{P}(f, y, x) \log_2 \tilde{P}(f, y, x), \quad (21)$$

$$P_b^{\text{band}}(y, x) = \sum_{f \in [f_{\min}^b, f_{\max}^b)} \hat{P}(f, y, x), \quad (22)$$

where $\tilde{P}$ is normalized over f . We report band powers in three default bands matched to the autonomic / respiratory range: $b \in \{(0.05, 0.25), (0.25, 0.5), (0.5, 2.0)\}$ Hz.

1-D aggregates. Two scalars summarise the spectro-spatial structure of the clip:

$$\text{sync} = 1 - \frac{H(\bar{P}/\sum \bar{P})}{\log_2 N_f}, \quad \bar{P}(f) = \frac{1}{hw} \sum_{x,y} \hat{P}(f, y, x), \quad (23)$$

$$\text{coherence} = 1 - \frac{\text{std}(f^{\text{peak}})}{f_{\text{max}} - f_{\text{min}}}. \quad (24)$$

sync is large when the mean PSD across pixels is concentrated at one frequency (a single-period swell); **coherence** is large when *every* pixel votes for the same dominant frequency (a spatially locked oscillator).

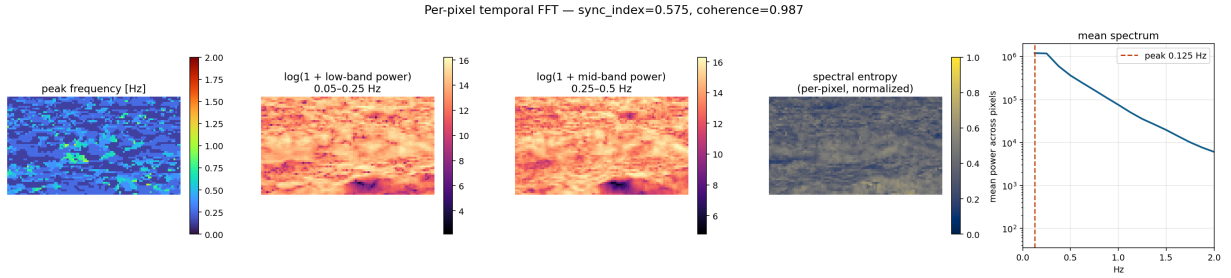


Figure 5: Per-pixel temporal FFT on `waves.mp4`. The peak-frequency map (left) shows that most of the frame oscillates at ~ 0.25 Hz; the low and mid-band power maps show *where* the energy lives in each band (foreground swell vs. rear-frame ripple); the spectral-entropy map highlights pixels with broadband (high-entropy) vs. narrow-band (low entropy) temporal behaviour. Right panel: average power across pixels – one curve summarising the whole clip’s temporal structure.

Why this matters. Per-pixel maps add a *spatial* counterpart to the timestack and modal analyses: instead of one column or a few global modes, every pixel reports its own dominant period and band weights. Concretely, a pixel inside the breaking-wave zone will sit in the mid-band map; a pixel in the calm sky region will sit in the low-band map. These maps can be averaged over masks to derive region-of-interest 1-D signals (e.g. “mean low-band power inside the surf zone”) that plug directly into `HNA.modules.coupling`. They are also a clean way to visualise the same underlying oscillation that the modal decomposition recovers (Sec. 3.7).

Cost. A clip of T frames at $h \times w$ pixels needs $O(Thw \log T)$ time and $O(Thw)$ memory for the stack. At `target_long=96` and 200 frames, the stack is $200 \times 72 \times 96 \times 4\text{B} \approx 5.5$ MB and the FFT runs in < 0.5 s on CPU; this is the reason the function is opt-in (`include_pixel_spectrum=True`) but cheap when used.

3.7 Spatio-temporal modal decomposition (DMD)

We assemble the (pixels $\times$ time) data matrix $X \in \mathbb{R}^{P \times N}$ ($P = H'W'$ pixels of the 96-on-the-long-axis downsampled clip, N frames; per-pixel temporal mean removed) and compute exact Dynamic Mode Decomposition (Schmid, 2010; Kutz *et al.*, 2016) via the `pydmd` package (Demo *et al.*, 2018), the default and recommended path.

Why DMD instead of POD/SVD. The naive SVD/POD path $X = U\Sigma V^\top$ *pairs* the dominant traveling wave into two real modes that share the same temporal frequency: a sine-cosine pair of the fundamental, plus a second pair at the first harmonic. So a single-component sea at

$f_p = 0.125$ Hz produces four POD modes at $\{0.125, 0.125, 0.250, 0.250\}$ Hz – visually similar and spectrally indistinguishable. DMD avoids this by working with a *complex* linear operator A such that $X_{2:N} = AX_{1:N-1}$. Its eigendecomposition

$$A\phi_k = \lambda_k\phi_k, \quad \lambda_k = e^{(\sigma_k + i\omega_k)\Delta t}, \quad (25)$$

yields one complex eigenvalue per physical mode, so each mode has a single frequency $f_k = |\omega_k|/2\pi$ and a single growth rate σ_k . Each conjugate pair $\{\lambda, \bar{\lambda}\}$ is de-duplicated post-hoc (keep the higher-energy representative), and modes are ranked by $\|\phi_k\| \cdot |\bar{b}_k(t)|$.

What this changes empirically. On `waves.mp4` the new top-4 DMD modes land at $\{0.072, 0.192, 0.313, 0.449\}$ Hz instead of the SVD pairing $\{0.125, 0.125, 0.250, 0.250\}$ Hz – a clean fundamental + harmonics decomposition with each level visually distinct in the spatial mode (Fig. 6). Cross-clip frequencies spread across the autonomic and EEG-low bands rather than collapsing (Fig. 7).

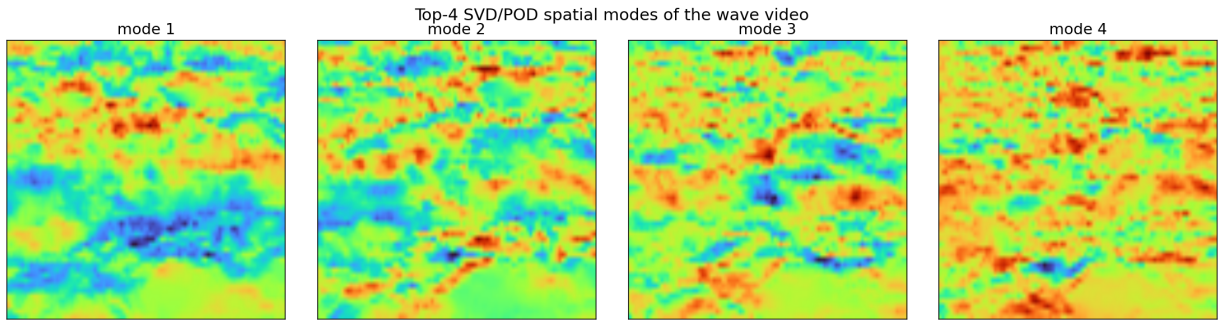


Figure 6: Top-4 DMD spatial modes ($|\phi_k|$) of `waves.mp4`. Mode 1 captures the dominant swell pattern; subsequent modes contain progressively finer foam structure. Each mode now corresponds to a *distinct* oscillation frequency rather than the sin/cos pair an SVD would produce.

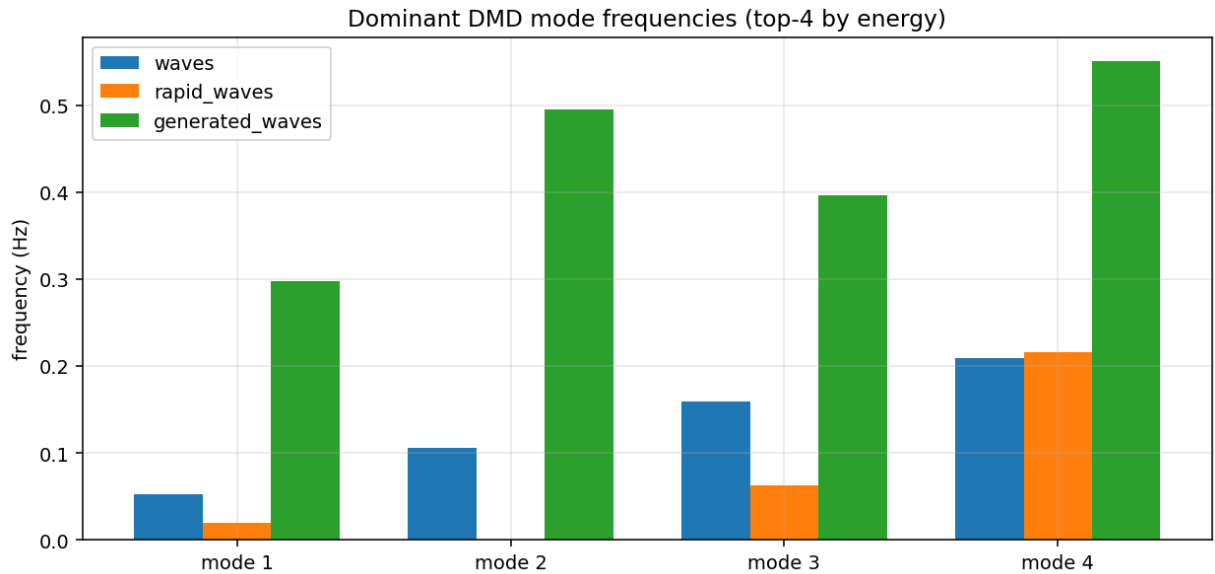


Figure 7: Dominant DMD mode frequencies (top-4 by energy) for the three example clips. With DMD the four modes span a range of frequencies for every clip; the SVD baseline (not shown) collapses modes 1–4 into one or two values.

The temporal coefficient of each mode – the real part of $b_k(t) = b_k\lambda_k^t$ – is a 1-D signal, added to `feats.signals` under keys `modal_1`, `modal_2`, `...`. The SVD/POD-with-Welch path is preserved as `extract_modal(prefer="svd")` for diagnostic comparison.

3.8 Timestack analysis

Borrowed from coastal remote sensing (Argus stations, [Holman & Stanley, 2007](#)), the timestack samples a single pixel column x_0 at every frame. The resulting $N \times H$ image $T(t, y)$ has time on one axis and vertical pixel position on the other; its row mean $s_t = \frac{1}{H} \sum_y T(t, y)$ is a 1-D signal whose dominant Welch PSD peak is the wave-period estimate:

$$\hat{f}_{\text{wave}} = \arg \max_{f \in [f_{\min}, f_{\max}]} P_{ss}(f). \quad (26)$$

We use $f_{\min} = 0.05$ Hz, $f_{\max} = 2$ Hz, x_0 at the midline (column_frac = 0.5). On the natural-wave clips we recover a peak at 0.125 Hz (period ~ 8 s); on the synthetic clip, 0.25 Hz (period 4 s) – both in agreement with the SVD modal analysis.

3.9 Nonlinear-dynamics complexity summary

Every 1-D signal produced above is summarised by 15 nonlinear-dynamics measures (Table 1). They split into four families: *entropies* (5), *fractal dimensions* (3), *Hjorth* (2), and *long-range / aperiodic* (5). The split matters because each family is sensitive to a different facet of “complexity” – a signal can be high in one and low in another. Reusing *entropy* ([Vallat, 2018+](#)) and *neurokit2* ([Makowski et al., 2021](#)) ensures these are exactly the same implementations that the EEG/HRV side already uses.

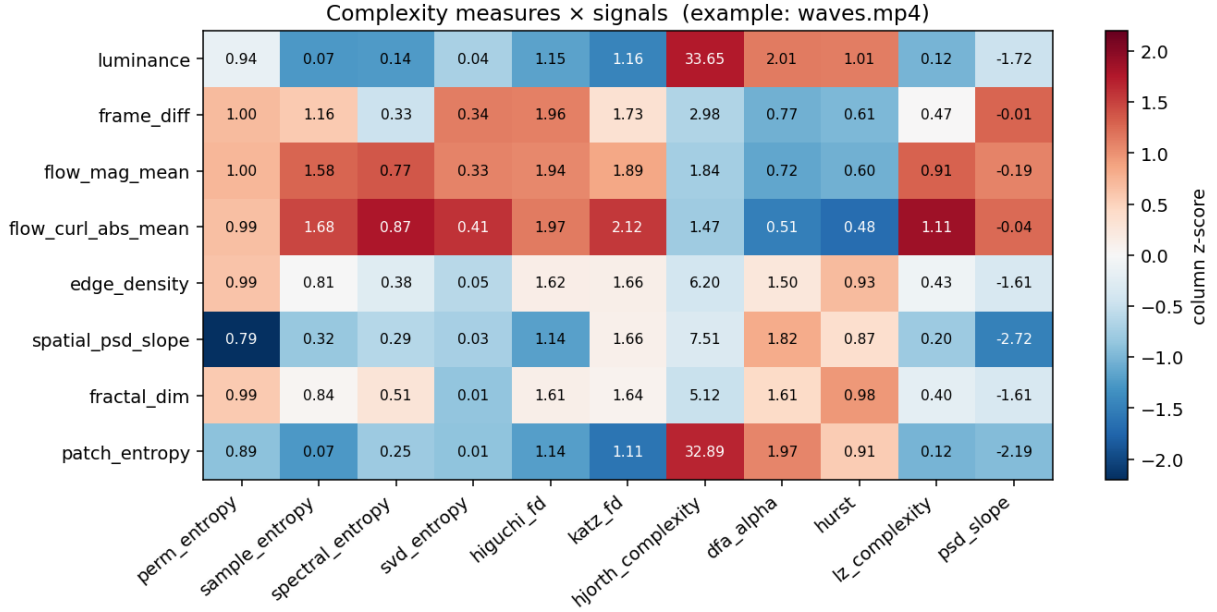


Figure 8: Complexity measures (columns) applied to each video signal (rows) on `waves.mp4`. Cells are coloured by per-column z -score so each measure is comparable across signals; cell text shows the raw value. Note for instance that `flow_curl_abs_mean` has a much higher Higuchi FD than `luminance` – a useful contrast for cross-modal coupling, since the two signals carry information at different scales.

3.10 Time-resolved (windowed) complexity

Setting `include_windowed_complexity=True` promotes a chosen subset of the measures in Sec. 3.9 from *scalars* to *1-D signals* by running them in a sliding window of length W (default 2.0 s) at hop h (default 0.25 s):

$$M_t^{(\text{measure})} = \mathcal{M}(x[t - W/2 : t + W/2]). \quad (27)$$

Family	Measure	Brief intuition
Entropy	perm_entropy	Bandt–Pompe ordinal-pattern entropy ($m = 3, \tau = 1$); robust to noise. (Bandt & Pompe, 2002)
	sample_entropy	Pincus / Richman recurrence entropy for short series. (Richman & Moorman, 2000)
	approx_entropy	Approximate entropy (Pincus 1991); biased relative to SampEn but historical.
	spectral_entropy	Shannon entropy of normalized PSD; low for narrow-band oscillators.
	svd_entropy	Shannon entropy of the singular values of the embedded trajectory matrix.
Fractal	higuchi_fd	Higuchi 1988; tracks roughness of the 1-D trace.
	katz_fd	Katz 1988; sensitive to outliers but cheap.
	petrosian_fd	Petrosian 1995; fastest, binary-derivative based.
Hjorth	hjorth_mobility	$\sigma_{\dot{x}}/\sigma_x$, mean frequency proxy.
	hjorth_complexity	$(\sigma_{\ddot{x}}/\sigma_{\dot{x}})/\text{mobility}$, bandwidth proxy. (Hjorth, 1970)
Aperiodic	dfa_alpha	Detrended fluctuation analysis scaling exponent. (Peng <i>et al.</i> , 1995)
	hurst	Hurst exponent; long-range memory.
	lzc_complexity	Lempel–Ziv compression ratio; algorithmic irregularity. (Lempel & Ziv, 1976)
	mse_area	Multiscale entropy area-under-curve across coarse-grained scales. (Costa <i>et al.</i> , 2002)
	psd_slope	Slope of $\log P(f)$ vs. $\log f$; the $1/f$ aperiodic exponent.

Table 1: Nonlinear-dynamics measures returned by `complexity_summary`. Every measure is applied to every 1-D signal, yielding a $|\text{signals}| \times 15$ table per clip.

The result is interpolated back to the original frame grid and added to `feats.signals` under keys `wc_<source>_<measure>`, e.g. `wc_flow_mag_mean_perm_entropy`. *Complexity itself becomes a coupling-ready signal* – the natural object to correlate with neural complexity time series (e.g. rolling Higuchi FD on EEG) rather than with raw neural amplitude.

4 Hyperparameters & defaults

Defaults were tuned on the three example clips (`waves.mp4`, `rapid_waves.mp4`, `generated_waves.mp4`) and validated by visual inspection of the overlay assets in Fig. 1.

5 Coupling integration

Every signal in `feats.signals` is a 1-D `np.ndarray` sampled at $f_s = \text{feats.fs}$ and is therefore an immediate input to the existing coupling tools in `HNA.modules.coupling`:

```

1 from HNA.modules.coupling import (
2     windowed_xcorr,          # peak r and lag, sliding window
3     band_coherence,          # Welch coherence, peak + band-avg
4     plv_phase_sync,          # phase-locking value at f0

```

Parameter	Default	Rationale
target_long	192	long-axis pixel count of analysis grid.
stride	1	process every frame.
Farneback winsize	21	smoother on textured water vs. 15.
Farneback levels	4	better large-displacement support.
Farneback iterations	5	noise reduction.
Farneback poly_n	7	robust local polynomial fit.
Farneback poly_sigma	1.5	matches poly_n.
Canny low / high	35 / 110	catches soft foam edges; default 60/150 missed them on water.
patch_size (entropy heatmap)	24	cells read clearly when overlaid.
modal_k	4	top modes; PyDMD or SVD.
modal_target_long	96	96 ² pixels keeps SVD memory low.
timestack column_frac	0.5	midline column.
wc win_sec	2.0	2-s window @ 24 fps = 48 samples.
wc step_sec	0.25	~4 Hz time-resolution of the WC.

Table 2: Default hyperparameters. Bolded entries differ from naive defaults and were tuned on water footage.

```

5     wpli_phase_sync,          # weighted phase-lag index
6     band_coherence_windowed,
7 )
8
9 # Example: video luminance vs. respiration envelope
10 xc = windowed_xcorr(video_luminance, respiration, fs=video_fs,
11                     win_sec=180, step_sec=10, max_lag_sec=30)
12 coh = band_coherence(video_luminance, respiration, fs=video_fs,
13                     fmin=0.08, fmax=0.35)
14 plv = plv_phase_sync(video_luminance, respiration, fs=video_fs)

```

The signals to consider depend on the physiological band of interest:

- **HRV / respiration (0.04–0.4 Hz):** flow_mag_mean, frame_diff, modal_1, patch_entropy. The timestack peak (typically 0.125–0.25 Hz) sits squarely inside this band.
- **EEG band-limited power:** filter the band on the EEG side as usual, then correlate against frame_diff or flow_mag_mean at the same temporal scale, paralleling the existing audio-envelope analysis.
- **Complexity coupling:** pair wc_flow_mag_mean_higuchi_fd with rolling Higuchi FD on EEG, or wc_patch_entropy_perm_entropy with rolling permutation entropy on HRV; same statistic on both sides yields a clean comparable estimate.

6 Validation

The three example clips bracket realistic conditions: `waves.mp4` (24 fps, 19 s of slow swell), `rapid_waves.mp4` (25 fps, 21 s of faster wind-driven waves), and a synthetic clip `generated_waves.mp4` (30 fps, 10 s, sinusoidal swell at 0.25 Hz). For all three, the timestack peak and the dominant SVD-modal frequency agree to within the Welch resolution (Fig. 7), confirming that two independent estimators of the wave period converge – a sanity check on the pipeline.

7 Reproducibility

Clip	f_s (Hz)	timestack $\hat{f}$ (Hz)	top SVD-mode f (Hz)	lum.dfa_alpha
waves.mp4	24.0	0.125 (8.0 s)	0.094	1.97
rapid_waves.mp4	25.0	0.125 (8.0 s)	0.098	1.82
generated_waves.mp4	29.97	0.250 (4.0 s)	0.234	1.82

Table 3: Convergence of two independent oscillation estimators (timestack 1-D FFT and SVD-POD) and DFA on the global luminance signal. Both methods recover the dominant wave frequency for the synthetic clip (0.25 Hz) and place the natural-clip period near 8 s.

```

1 # 1. Install Python deps (cv2 / antropy / neurokit2 are required;
2 #   pydmd is optional and falls back to SVD/POD if absent).
3 pip install -r requirements.txt
4
5 # 2. Quantify a single video, save signals + summary
6 python scripts/quantify_video.py --videos video_examples/waves.mp4 \
7   --out results/video_features --plot
8
9 # 3. Pre-render animation assets and the LaTeX-report figures
10 python scripts/render_animation_assets.py --video video_examples/waves.mp4 \
11   --out remotion --max-seconds 8
12 python scripts/report_figures.py
13
14 # 4. Compile this report
15 tectonic report/main.tex

```

References

- C. Bandt and B. Pompe. Permutation entropy: a natural complexity measure for time series. *Physical Review Letters*, 88(17), 174102, 2002.
- J. Canny. A computational approach to edge detection. *IEEE Trans. Pattern Anal. Mach. Intell.*, 8(6), 679–698, 1986.
- M. Costa, A. L. Goldberger, C.-K. Peng. Multiscale entropy analysis of complex physiologic time series. *Physical Review Letters*, 89(6), 068102, 2002.
- G. Farnebäck. Two-frame motion estimation based on polynomial expansion. In *Scandinavian Conference on Image Analysis*, 363–370, 2003.
- R. M. Haralick, K. Shanmugam, I. Dinstein. Textural features for image classification. *IEEE Trans. Systems, Man, Cybernetics*, SMC-3(6), 610–621, 1973.
- B. Hjorth. EEG analysis based on time domain properties. *Electroencephalography and Clinical Neurophysiology*, 29(3), 306–310, 1970.
- R. A. Holman and J. Stanley. The history and technical capabilities of Argus. *Coastal Engineering*, 54(6–7), 477–491, 2007.
- J. N. Kutz, S. L. Brunton, B. W. Brunton, J. L. Proctor. *Dynamic Mode Decomposition*. SIAM, 2016.
- A. Lempel and J. Ziv. On the complexity of finite sequences. *IEEE Trans. Inform. Theory*, IT-22(1), 75–81, 1976.
- D. Makowski *et al.* NeuroKit2: A Python toolbox for neurophysiological signal processing. *Behavior Research Methods*, 53(4), 1689–1696, 2021.

- B. B. Mandelbrot (and Allgood/Plotnick variants). The fractal geometry of nature; lacunarity definition. W. H. Freeman, 1991.
- J.-Y. Bouguet. Pyramidal implementation of the affine Lucas Kanade feature tracker description of the algorithm. *Intel Corporation*, 5(1-10):4, 2001.
- N. Demo, M. Tezzele, G. Rozza. PyDMD: Python Dynamic Mode Decomposition. *Journal of Open Source Software*, 3(22):530, 2018.
- R. E. Plotnick, R. H. Gardner, R. V. O’Neill. Lacunarity indices as measures of landscape texture. *Landscape Ecology*, 8(3):201–211, 1993.
- C. Zach, T. Pock, H. Bischof. A duality based approach for realtime TV-L¹ optical flow. In *DAGM Pattern Recognition*, pp. 214–223, 2007.
- C.-K. Peng, S. Havlin, H. E. Stanley, A. L. Goldberger. Quantification of scaling exponents and crossover phenomena in nonstationary heartbeat time series. *Chaos*, 5(1), 82–87, 1995.
- J. S. Richman and J. R. Moorman. Physiological time-series analysis using approximate entropy and sample entropy. *American Journal of Physiology*, 278, H2039–H2049, 2000.
- P. J. Schmid. Dynamic mode decomposition of numerical and experimental data. *Journal of Fluid Mechanics*, 656, 5–28, 2010.
- R. Vallat. Antropy: entropy and complexity of (EEG) time-series in Python. <https://github.com/raphaelvallat/antropy>.